ACN QUIZ-VIVA

Created Using Chat-GPT

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📘 Experiment 1: PC-to-PC Network (Crossover Cable)

◆ Viva Questions (Q&A)

1. **What is a crossover cable?**
   → A cable used to directly connect two similar devices by crossing Tx/Rx pairs.
2. **Why is a crossover cable required?**
   → Because both PCs use same pin configuration.
3. **What layer is involved here?**
   → Physical + Data Link Layer.
4. **What is peer-to-peer networking?**
   → Direct communication without central device.
5. **What happens if IPs differ in subnet?**
   → Communication fails.
6. **Which command tests connectivity?**
   → `ping`
7. **Is default gateway required?**
   → No
8. **Which port is used?**
   → FastEthernet
9. **What protocol is used in ping?**
   → ICMP
10. **Can switch replace crossover?**
    → Yes (with straight cable)

◆ Fill in the Blanks

1. Crossover cable is used to connect two _____ devices. (**similar**)
2. IPs must be in same _____ for communication. (**network**)
3. Ping uses _____ protocol. (**ICMP**)
4. No _____ device is required. (**intermediate**)
5. Communication is _____ type. (**peer-to-peer**)

◆ MCQ

1. Crossover cable is used for:
   a) Router-PC
   b) PC-PC ✅
   c) Switch-PC
   d) Router-Router
2. Ping uses:
   a) TCP
   b) UDP
   c) ICMP ✅
   d) ARP
3. Required IP condition:
   a) Different network
   b) Same network ✅
   c) Random
   d) Public IP
4. Layer involved:
   a) Application
   b) Physical ✅
   c) Session
   d) Transport
5. Gateway required?
   a) Yes
   b) No ✅

◆ True/False

1. Crossover cable connects similar devices → **True**
2. Gateway is required → **False**
3. Ping uses ICMP → **True**
4. Different subnet works → **False**
5. Switch is mandatory → **False**

📘 Experiment 2: LAN using Hub & Switch

◆ Viva (10)

1. What is LAN? → Local network
2. Hub works at? → Physical layer
3. Switch works at? → Data Link layer
4. What is MAC table? → Address mapping table
5. Hub behavior? → Broadcast
6. Switch behavior? → Forward selectively
7. Collision domain in hub? → Single
8. In switch? → Multiple
9. Which is smarter? → Switch
10. Why? → MAC-based forwarding

◆ Fill in the Blanks

1. Hub works at _____ layer. (**Physical**)
2. Switch uses _____ address. (**MAC**)
3. Hub creates _____ domain. (**single collision**)
4. Switch reduces _____. (**traffic**)
5. LAN stands for _____. (**Local Area Network**)

◆ MCQ

1. Hub sends data to:
   a) One
   b) All ✅
   c) Router
   d) None
2. Switch uses:
   a) IP
   b) MAC ✅
   c) Port
   d) DNS
3. Collision domain (hub):
   a) Many
   b) One ✅

4. Smart device:
   a) Hub
   b) Switch ✅
5. Layer:
   a) Switch → L2 ✅

◆ True/False

1. Hub is intelligent → False
2. Switch uses MAC → True
3. Hub reduces traffic → False
4. Switch reduces collision → True
5. LAN is limited area → True

📘 Experiment 3: Router & Default Gateway

◆ Viva

1. What is router? → Network connector
2. Default gateway? → Exit point
3. Layer? → Network
4. Routing function? → Packet forwarding
5. Gateway required? → Yes
6. Protocol used? → IP
7. Without gateway? → No external comm
8. Router uses? → Routing table
9. Interface config? → Required
10. Ping used? → Yes

◆ Fill

1. Router works at _____ layer (**Network**)
2. Gateway is _____ point (**exit**)
3. Routing uses _____ (**IP**)
4. Ping checks _____ (**connectivity**)
5. Router connects _____ networks (**multiple**)

◆ MCQ

1. Router layer: Network ✅
2. Gateway needed: Yes ✅
3. Router uses: IP ✅
4. Device type: Layer 3 ✅
5. Protocol: IP ✅

◆ True/False

1. Router connects networks → True
2. Gateway optional → False
3. Routing automatic → False
4. IP required → True
5. Ping tests connectivity → True

📘 Experiment 4: Static Routing

◆ Viva

1. Static routing? → Manual
2. Command? → `ip route`
3. Advantage? → Control
4. Disadvantage? → No auto update
5. Next-hop? → Router IP
6. Routing table? → Manual
7. Used where? → Small network
8. Protocol? → None
9. Dynamic? → No
10. Reliability? → High control

◆ Fill

1. Static routing is _____ (**manual**)
2. Command is _____ (**ip route**)
3. Uses _____ IP (**next-hop**)
4. No _____ update (**automatic**)
5. Table updated _____ (**manually**)

◆ MCQ

1. Command: ip route ✅
2. Type: Manual ✅
3. Uses: Next hop ✅
4. Dynamic? No ✅
5. Best for: Small network ✅

◆ True/False

1. Static routing automatic → False
2. Uses ip route → True
3. Needs admin → True
4. Dynamic update → False
5. Predictable → True

📘 Experiment 5: RIP Routing

◆ Viva

1. RIP type? → Distance vector
2. Metric? → Hop count
3. Max hop? → 15
4. Command? → `router rip`
5. Version? → v2
6. Update? → Periodic
7. Auto? → Yes

8. Protocol? → UDP
9. Port? → 520
10. Convergence? → Fast

◆ Fill

1. RIP uses _____ (**hop count**)
2. Max hop _____ (**15**)
3. Protocol _____ (**UDP**)
4. Port _____ (**520**)
5. Command _____ (**router rip**)

◆ MCQ

1. RIP metric: Hop ✅
2. Max hop: 15 ✅
3. Type: Distance vector ✅
4. Protocol: UDP ✅
5. Port: 520 ✅

◆ True/False

1. RIP dynamic → True
2. Hop >15 allowed → False
3. Uses UDP → True
4. Static routing → False
5. Auto update → True

📄 Experiment 6: ACL Configuration

◆ Viva (IMPORTANT)

1. ACL purpose? → Traffic filtering
2. Standard ACL? → Source only
3. Extended ACL? → Source+Dest+Protocol
4. Command? → `access-list`
5. Apply? → `ip access-group`
6. Direction? → in/out
7. Processing? → Top-down
8. Default rule? → Deny all
9. Number range? → 1–99
10. Wildcard? → 0.0.0.255

◆ Fill

1. ACL filters _____ (**traffic**)
2. Standard uses _____ (**source IP**)
3. Apply using _____ (**ip access-group**)
4. Default action _____ (**deny**)
5. Wildcard _____ (**inverse mask**)

◆ MCQ

1. ACL command: access-list ✅
2. Apply: ip access-group ✅
3. Standard uses: Source ✅
4. Default: Deny ✅
5. Order: Top-down ✅

◆ True/False

1. ACL default permit → False
2. ACL filters traffic → True
3. Extended uses protocol → True
4. Order not important → False
5. Deny all at end → True